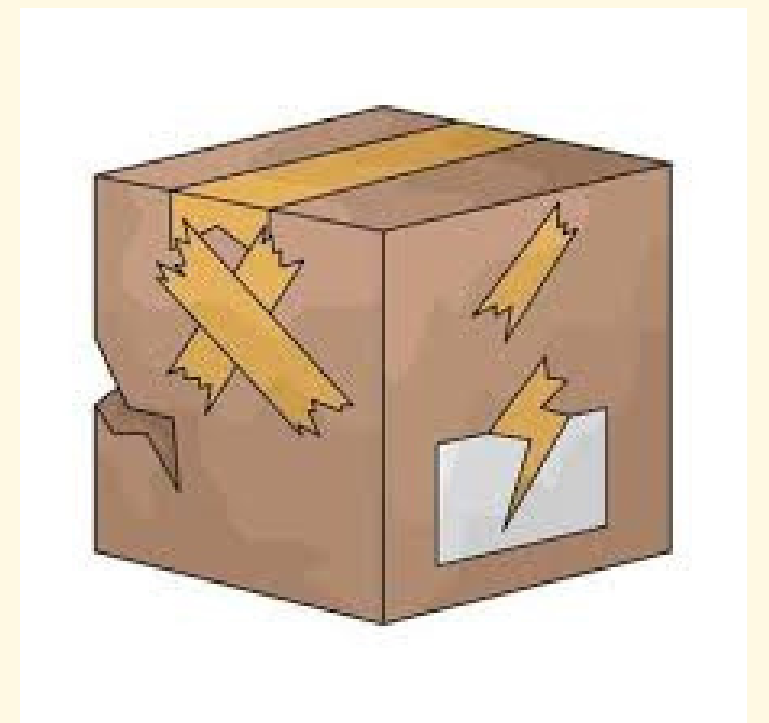
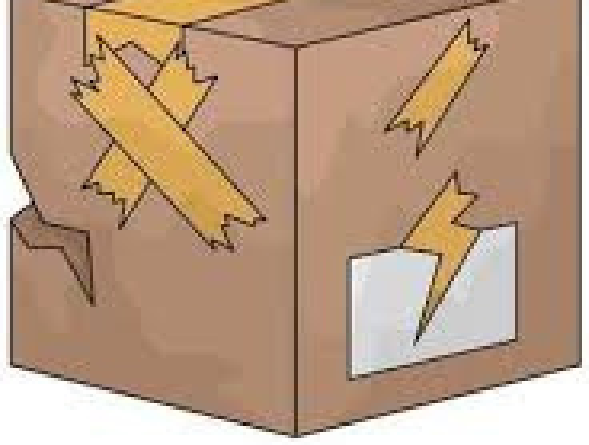




Automated Parcel Damage Detection System for E-Commerce Fulfillment Centres





Introduction

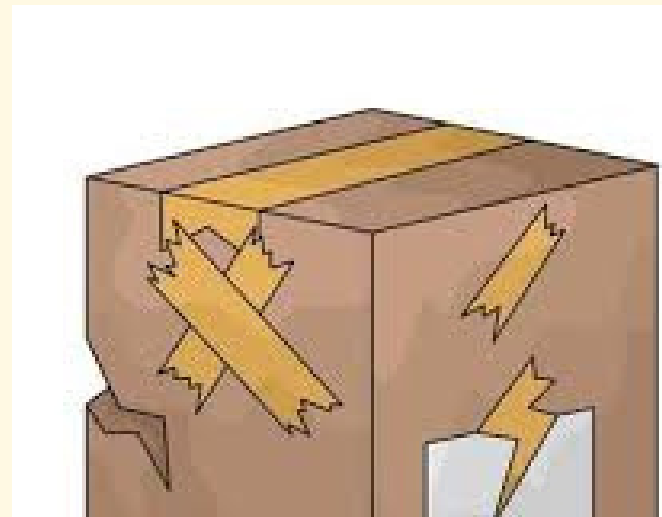
The rapid growth of e-commerce has increased the number of parcels processed in fulfillment centers every day. During handling and transportation, packages may get damaged due to improper stacking, impacts, or environmental conditions. Manual inspection is often slow and inaccurate, making it difficult to detect damaged parcels efficiently. This project proposes an Automated Parcel Damage Detection System that uses computer vision and deep learning to identify damaged packages in real time, improving quality control and reducing customer complaints.





Problem Statement

E-commerce fulfillment centers process thousands of parcels daily, making manual inspection difficult and unreliable. Human inspectors may miss damaged packages due to fatigue and high conveyor belt speeds, resulting in damaged products reaching customers. This leads to increased returns, financial losses, and reduced customer satisfaction. Therefore, an automated system is needed to accurately detect parcel damage in real time and support efficient quality control.





Objectives

- To automatically detect damaged parcels using AI and computer vision.
- To identify different damage types such as crushed, wet, puncture, dirt, and tear.
- To reduce manual inspection efforts and human errors.
- To classify damage based on severity levels.
- To improve operational efficiency and customer satisfaction.

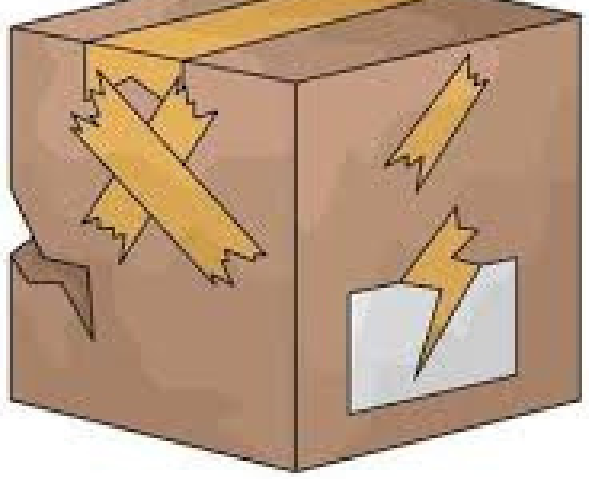




Challenges Faced

- Limited dataset with only a small number of training images.
- Imbalance in damage classes, causing some damages to have fewer samples.
- Lack of real conveyor-belt images and camera hardware.
- Low initial model accuracy during training.
- Integration of detection results with Power BI analytics.
- Real-time processing requirements for fast-moving parcels.





What is Real-Time Deployment?

Real-time deployment refers to the process of detecting parcel damage instantly as packages move through the fulfillment center. Since a physical camera is not available, this project simulates real-time operation by processing parcel images from a folder one by one, similar to a live conveyor belt camera feed.



Deployment Process

1. Image Input

Parcel images are stored in a dataset folder.

Images are loaded sequentially to simulate conveyor belt movement.

2. Preprocessing

Images are resized to 640×640 pixels using OpenCV.

Improves consistency for model prediction.

3. Damage Detection

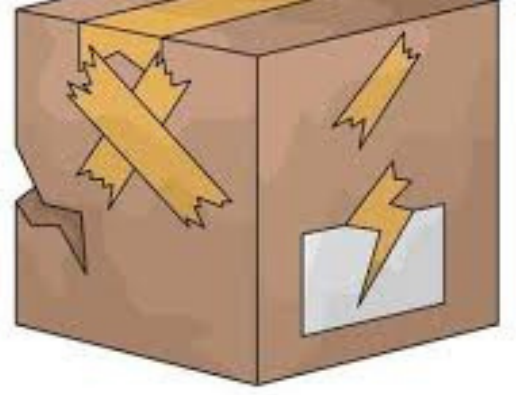
RF-DETR/YOLOv8 model analyzes each image.

Detects damage type, confidence score, and location.

4. Severity Classification

Damage is categorized into four severity levels.

Appropriate action is assigned automatically.



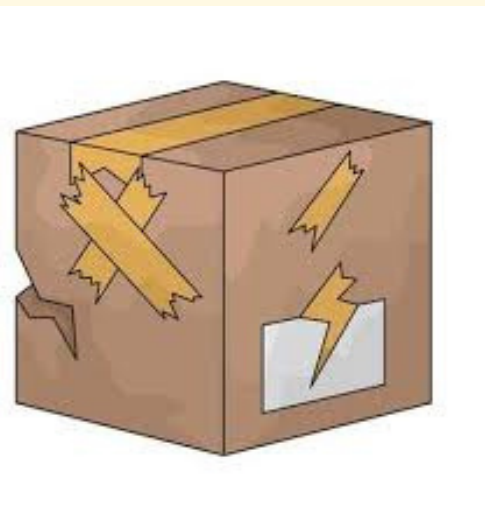
Damage Type	Severity	Action
Wet, Dirt	Minor	Log Only
Tear/Torn	Moderate	Review
Puncture	Major	QC Inspection
Crushed	Critical	Reject & Alert



5. Logging and Analytics

Detection results are stored automatically.

Data is converted into reports and displayed in Power BI dashboards.

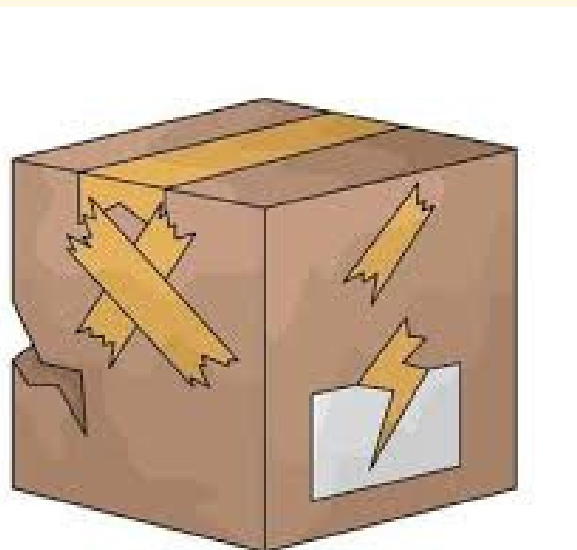


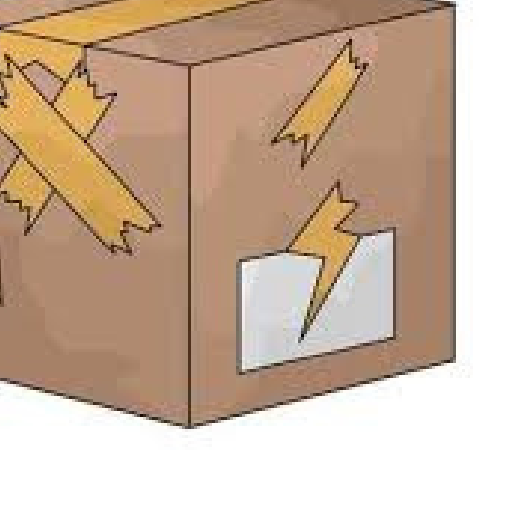


Results and Analysis

Model Performance

The trained RF-DETR/YOLOv8 model was evaluated on the parcel damage dataset to measure its detection accuracy and reliability. Data augmentation techniques significantly improved the model's performance.





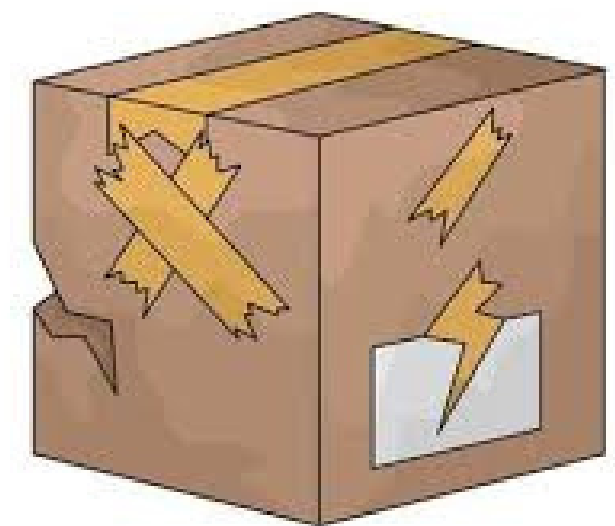
Metric	Before Augmentation	After Augmentation
mAP@50	269%	45–55%
Precision	305%	50–60%
Recall	391%	55–65%
F1-Score	343%	50–60%





Key Results

- Successfully detected 5 types of parcel damage.
- Improved accuracy through data augmentation.
- Performed real-time damage detection on simulated conveyor belt images.
- Automatically classified damage severity levels.
- Generated logs and analytics for Power BI dashboards.
- Reduced dependency on manual inspection.





Thank You

